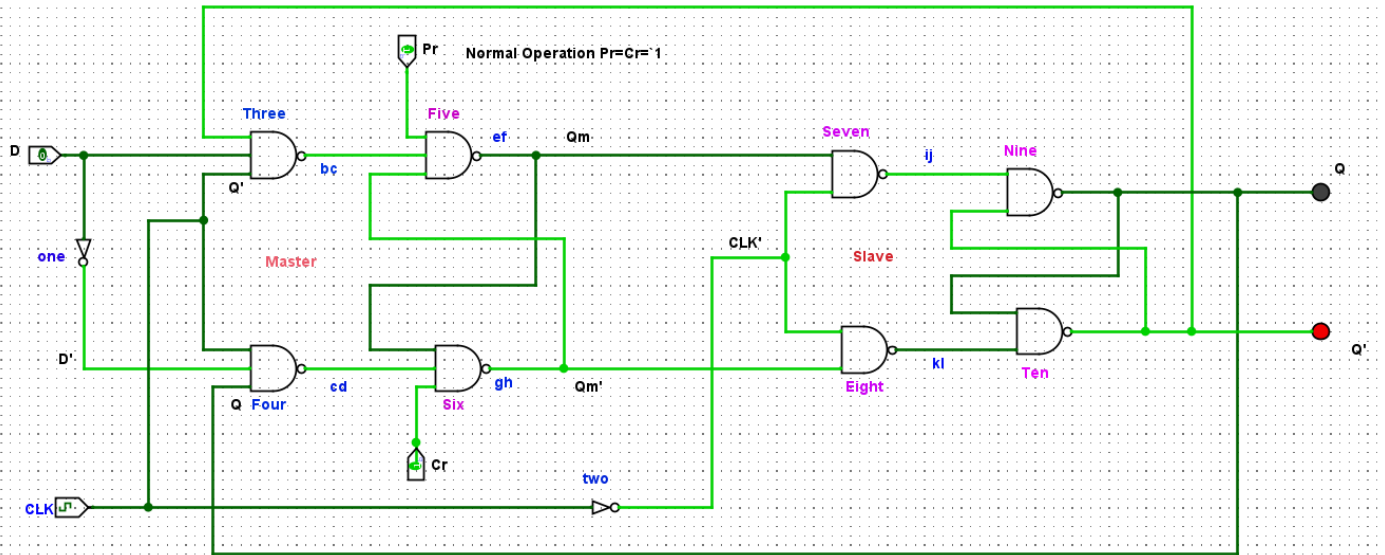
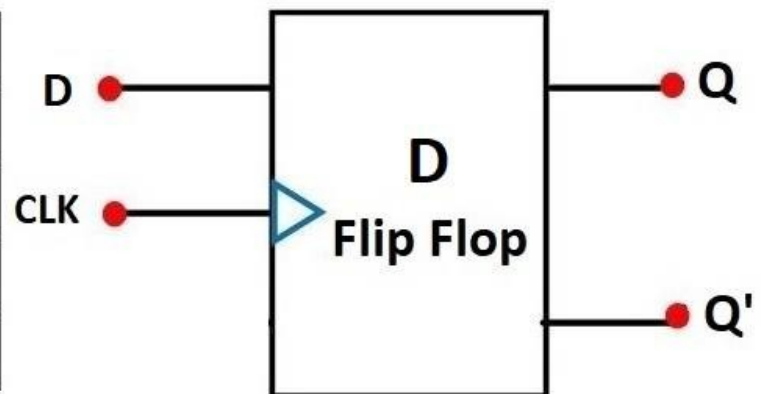




Truth Table of DFlip Flop

D	CLK	Q	Q'
0	0	0	1
0	1	0	1
1	0	0	1
1	1	1	0



```

`timescale 1ns/1ps

module d_type (
input wire clk,
input wire d,
input pr,
input cr,
output wire q,
output wire q_bar );

wire d_bar, clk_bar;
wire bc,cd,ef,gh, ij,kl;

not #1 one (d_bar, d);
not #1 two (clk_bar, clk);

nand #1 three (bc, d, clk, q_bar);
nand #1 four (cd, d_bar, clk, q);

nand #1 five (ef, pr, bc, gh);
nand #1 six (gh, cd, cr, ef);

nand #1 seven (ij, ef, clk_bar);
nand #1 eight (kl, gh, clk_bar);

nand #1 nine (q, q_bar, ij);
nand #1 ten (q_bar, kl, q);

endmodule

```

```

`timescale 1ns/1ps

module d_type_tb;

    reg clk, d, pr, cr;
    wire q, q_bar;

    d_type dut (
        .clk(clk), .d(d), .pr(pr), .cr(cr),
        .q(q), .q_bar(q_bar)
    );

    initial clk = 0;
    always #5 clk = ~clk;

    initial begin
        $dumpfile("d_type_tb.vcd");
        $dumpvars(0, d_type_tb);

        d = 0;

        pr = 1; cr = 1; #10;    // normal
        pr = 0; cr = 1; #10;    // preset
        pr = 1; cr = 0; #10;    // clear
        pr = 1; cr = 1; #10;    // back to normal

        repeat (10) begin
            d = $random;
            #10;
        end

        #10 $finish;
    end

endmodule

```

